CREDIFY

Credit Card Platform — Product Requirements Document

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01 Product Vision & Goals

Problem Statement

Millions of credit card users in emerging and developed markets face a fragmented experience: outdated bank apps, opaque billing cycles, lack of real-time spend intelligence, and poor customer support. Existing solutions are either too rigid (legacy banking apps) or too narrow (neobank point solutions) to deliver a unified, intelligent credit card experience.

Credify solves this by providing a modern, API-first credit card platform with intelligent spend analytics, seamless payments, real-time notifications, and an embedded credit scoring engine — all within a single mobile-first product.

Target Users (Personas)

The Young Professional	**Age 24–35, salaried, urban**	**No spend visibility, surprise bills**	**Budget control + rewards**
The Small Business Owner	**Age 30–50, self-employed**	**Expense tracking, cash flow**	**Separate biz/personal spend**
The Credit Builder	**Age 20–28, thin credit file**	**No credit history, rejections**	**Build score responsibly**
The Power Spender	**Age 35–55, high income**	**Complex rewards, max utilization**	**Optimize rewards & cashback**

Core Value Proposition

- Real-time spend intelligence with category-level insights
- Intelligent credit scoring engine with transparent score factors
- Rewards & cashback with personalized redemption pathways
- Instant payments with multi-payment-method support
- 24/7 AI-powered support with human escalation
- Bank-grade security: tokenization, biometrics, fraud alerts

Market Differentiation

			✅
Real-time analytics	❌ **No**	**Partial**	✅ **Full**
Credit score engine	❌ **3rd party**	❌ **No**	✅ **In-house**
Custom rewards engine	**Partial**	**Basic**	✅ **Advanced**
API-first architecture	❌ **No**	**Partial**	✅ **Full**
KYC + AML built-in	✅ **Yes**	**Partial**	✅ **Yes**
ML Fraud detection	**Partial**	**Partial**	✅ **Yes**

02 Feature Breakdown (Modular)

Module 1 — User & Auth (with KYC)

Description

Handles user onboarding, authentication, identity verification, and session management. Core of the platform — every other module depends on this.

Key Features

- Phone/email OTP + biometric login (Face ID / Fingerprint)

- JWT-based auth with refresh token rotation
- KYC: Aadhaar/PAN verification (India), SSN/Driver License (US)
- Liveness detection + document OCR via third-party (Onfido / DigiLocker)
- Role-based access: Customer, Agent, Admin, Super Admin
- Device trust management + suspicious login alerting

User Flow

- User downloads app → enters phone number
- OTP sent via SMS → user verifies
- Basic profile created (name, DOB, email)
- KYC initiated → upload ID docs → liveness check
- KYC reviewed (automated + manual fallback)
- Account activated → credit assessment triggered

Edge Cases

- KYC failure → manual review queue, 48hr SLA
- Duplicate identity detected → block + flag for compliance
- Biometric unavailable → PIN fallback
- Session expired → graceful re-auth without data loss

Module 2 — Credit Card Management

Key Features

- Virtual card instant issuance post KYC approval
- Physical card request with delivery tracking
- Card freeze/unfreeze with one tap
- Credit limit management (view, request increase)
- Spending controls: merchant category blocks, geo-restrictions
- Card replacement flow (lost/stolen)
- Multi-card support (primary + supplementary)

User Flow

- Credit limit assigned by scoring engine
- Virtual card issued immediately with card number, CVV, expiry
- User sets PIN → card is live
- Physical card dispatched (T+3 to T+7)
- Activate physical card via app → virtual deprecated

Edge Cases

- Card expired → auto-renewal 30 days before expiry
- CVV/card number leaked → instant invalidation + reissue
- Limit utilization > 90% → proactive alert to user

Module 3 — Transactions

Key Features

- Real-time transaction push (webhooks from card network)
- Merchant enrichment: logo, category, location from Clearbit / Plaid
- Transaction search, filter, sort
- Receipt attachment (photo upload)
- Split transactions across categories
- Dispute initiation with evidence upload
- International transactions with FX rate display

Edge Cases

- Duplicate transaction detected → auto-flag, hold + user alert
- Offline merchant data → fallback to raw merchant string
- Large transaction (> 5x avg) → fraud check + confirmation
- Disputed amount > 30 days old → rejection with explanation

Module 4 — Billing & Payments

Key Features

- Monthly statement generation (PDF + in-app)
- Minimum payment, full payment, custom amount
- Auto-pay setup with configurable triggers
- Multi-payment methods: UPI, NEFT, debit card, net banking
- Due date reminders (3 days, 1 day, day-of)
- Late fee waiver request (CRM-linked)
- Installment conversion (EMI) for eligible transactions

Billing Cycle

- Statement generated on billing date (e.g., 1st of month)
- Payment due date = billing date + 20 days
- User receives statement via email + push notification
- Payment window open → user pays any amount
- Interest computed on remaining balance after due date

Edge Cases

- Payment gateway failure → retry queue with exponential backoff
- Partial payment → outstanding balance carries with interest
- Overlimit payment → excess refunded within 3 business days

Module 5 — Rewards

Key Features

- Earn points on every transaction (configurable multipliers per category)
- Bonus rewards: merchant partnerships, seasonal campaigns
- Reward redemption: cashback, vouchers, flight miles, merchandise
- Points expiry management with 90-day pre-expiry alerts
- Referral rewards program
- Tiered membership (Silver → Gold → Platinum)

Rewards Engine Logic

Base: 1 point per ₹100 spent | Dining: 5x | Travel: 3x | Fuel: 1x (capped) | Partner merchants: up to 10x

Edge Cases

- Reversed transaction → points auto-deducted
- Points fraud (fake referrals) → velocity checks
- Expired points → non-reversible, user notified in advance

Module 6 — Notifications

Key Features

- Push, SMS, Email, WhatsApp channels
- Transaction alerts (real-time)
- Payment reminders (scheduled)
- Fraud alerts (immediate)
- Reward updates, offer notifications
- User-configurable notification preferences

Notification Rules Engine

Event-driven rules: IF transaction > ₹10,000 → SMS + Push | IF due date T-3 → Email + Push | IF fraud score > 0.8 → Immediate multi-channel alert

Module 7 — CRM / Support

Key Features

- In-app chat with AI-first response (Claude/GPT integration)
- Ticket creation with category tagging

- Human escalation with full context handoff
- Callback scheduling
- Knowledge base / FAQ search
- Complaint tracking with SLA enforcement
- CSAT collection post-resolution

Edge Cases

- AI unable to resolve → auto-escalate with priority flag
- Multiple open tickets for same issue → deduplicate
- VIP customer (Platinum) → priority queue routing

Module 8 — Analytics & Insights

Key Features

- Spend breakdown by category (pie/bar charts)
- Monthly vs previous month comparison
- Merchant-level spend history
- Budget setting and tracking
- Cashback / rewards earned this month
- Credit utilization trend
- Upcoming bill predictions based on recurring spend

Module 9 — Credit Scoring Engine

Inputs

- Payment history (on-time vs late)
- Credit utilization ratio
- Account age and credit mix
- Hard/soft inquiry history
- Bureau data (CIBIL/Experian integration)

Output

- Internal Credify Score (300–900)
- Score change factors (what improved / worsened)
- Limit recommendation signal
- Risk band: Low / Medium / High / Very High

03 System Architecture

High-Level Architecture

```
=== CREDIFY — HIGH-LEVEL SYSTEM ARCHITECTURE ===

┌──────────────────────────────────────────────────────────────────────┐
│                              CLIENT LAYER                            │
│   [Mobile App iOS]   [Mobile App Android]   [Web Dashboard]          │
└──────────────────────────────────────────────────────────────────────┘
                              │ HTTPS / WSS
                              ▼
┌──────────────────────────────────────────────────────────────────────┐
│                   API GATEWAY (Kong / AWS API GW)                    │
│   Rate Limiting │ Auth Middleware │ SSL Termination │ Logging        │
└──────────────────────────────────────────────────────────────────────┘
      │            │            │            │            │
      ▼            ▼            ▼            ▼            ▼
 ┌─────────┐   ┌─────────┐   ┌─────────┐   ┌─────────┐   ┌─────────┐
 │ Auth    │   │ Card    │   │ Txn     │   │ Billing │   │ Rewards │
 │ Service │   │ Service │   │ Service │   │ Service │   │ Service │
 └─────────┘   └─────────┘   └─────────┘   └─────────┘   └─────────┘
      │            │            │            │            │
      ▼            ▼            ▼            ▼            ▼
┌──────────────────────────────────────────────────────────────────────┐
│                       MESSAGE BUS (Kafka / RabbitMQ)                 │
│          Events: txn.created │ payment.received │ fraud.alert        │
└──────────────────────────────────────────────────────────────────────┘
      │            │            │            │            │
      ▼            ▼            ▼            ▼            ▼
 ┌─────────┐   ┌─────────┐   ┌─────────┐   ┌─────────┐   ┌─────────┐
 │ Notif   │   │ CRM     │   │ Score   │   │ Fraud   │   │ Analytics │
 │ Service │   │ Service │   │ Engine  │   │ Engine  │   │ Service │
 └─────────┘   └─────────┘   └─────────┘   └─────────┘   └─────────┘

┌──────────────────────────────────────────────────────────────────────┐
│                              DATA LAYER                              │
│   PostgreSQL (Primary) │ Redis (Cache) │ S3 (Docs/Media)             │
│   ClickHouse (Analytics) │ Elasticsearch (Search/Logs)               │
└──────────────────────────────────────────────────────────────────────┘

┌──────────────────────────────────────────────────────────────────────┐
│                         EXTERNAL INTEGRATIONS                        │
│   Card Network (Visa/MC) │ Bureau (CIBIL) │ KYC (Onfido)             │
│   Payment GW (Razorpay)  │ SMS (Twilio)   │ Email (SendGrid)         │
└──────────────────────────────────────────────────────────────────────┘
```

Service Breakdown

API Gateway	**Kong / AWS API GW**	**Routing, rate limiting, auth middleware, SSL**
Auth Service	**Django REST + JWT**	**OTP, KYC flow, session management, RBAC**
Card Service	**Django REST**	**Card issuance, controls, limit management**
Transaction Service	**Django REST + Celery**	**Real-time txn ingestion, enrichment, dispute**
Billing Service	**Django REST + Celery Beat**	**Statement gen, payment processing, EMI**
Rewards Service	**Django REST**	**Points engine, tiers, redemption, campaigns**
Notification Service	**FastAPI + Celery**	**Rules engine, multi-channel dispatch**

CRM Service	Django REST	Ticket management, AI chat, SLA tracking
Scoring Engine	FastAPI + scikit-learn	Credit score computation, risk banding
Fraud Engine	FastAPI + ML models	Real-time fraud scoring, velocity checks
Analytics Service	Django + ClickHouse	Spend insights, admin dashboard, KPIs

Async Processing & Event Flow

All inter-service communication for non-blocking operations uses Kafka topics:

```
KAFKA TOPICS:

user.kyc.completed       → triggers: Card issuance, Score init
txn.created              → triggers: Notification, Rewards, Fraud check
txn.disputed             → triggers: CRM ticket creation
payment.received         → triggers: Billing update, Notification
fraud.alert.raised       → triggers: Card freeze, Notification (immediate)
reward.points.expiring   → triggers: Notification (T-90 days)
score.updated            → triggers: Limit review signal

CELERY BEAT SCHEDULED JOBS:
statement_generation     → runs on billing date (monthly)
due_date_reminder        → runs daily at 9AM
points_expiry_check      → runs weekly
score_refresh            → runs monthly per user
```

04 Database Design

Core Tables

users	id, phone, email, name, dob, status, kyc_status, created_at	1:1 → kyc_records, 1:M → cards
kyc_records	id, user_id, doc_type, doc_number, verified_at, provider_ref, status	M:1 → users
cards	id, user_id, card_number_token, cvv_token, expiry, status, credit_limit, available_limit, type	M:1 → users; 1:M → transactions
transactions	id, card_id, amount, currency, merchant_name, merchant_category, status, txn_type, posted_at, auth_code	M:1 → cards; 1:M → disputes

billing_cycles	**id, user_id, start_date, end_date, total_spend, min_due, total_due, statement_date, status**	**M:1 → users; 1:M → payments**
payments	**id, billing_cycle_id, user_id, amount, method, gateway_ref, status, paid_at**	**M:1 → billing_cycles**
rewards_ledger	**id, user_id, txn_id, points_earned, points_redeemed, event_type, balance_after, created_at**	**M:1 → users, transactions**
notifications	**id, user_id, channel, template_key, payload, status, sent_at, read_at**	**M:1 → users**
support_tickets	**id, user_id, category, subject, status, priority, assigned_agent_id, resolved_at, csat_score**	**M:1 → users**
credit_scores	**id, user_id, score, risk_band, computed_at, factors_json, bureau_pull_id**	**M:1 → users**
fraud_alerts	**id, txn_id, user_id, score, rule_triggered, action_taken, reviewed_at**	**M:1 → transactions**
audit_logs	**id, entity_type, entity_id, action, actor_id, old_value, new_value, ip_address, created_at**	**Polymorphic**

ER-Level Relationships

User (1) → (M) Cards → (M) Transactions → (M) Disputes

User (1) → (M) BillingCycles → (M) Payments

User (1) → (M) RewardsLedger entries (linked to Transactions)

User (1) → (M) CreditScores (versioned, one active at a time)

User (1) → (M) SupportTickets

Transaction (1) → (1) FraudAlert (when triggered)

All entities → (M) AuditLogs via polymorphic entity_type + entity_id

05 API Design (Django REST Framework)

Auth Module

POST	/api/v1/auth/request-otp/	Send OTP to phone	Public
POST	/api/v1/auth/verify-otp/	Verify OTP → issue JWT	Public
POST	/api/v1/auth/token/refresh/	Rotate refresh token	Refresh token
POST	/api/v1/auth/kyc/initiate/	Start KYC session	JWT
GET	/api/v1/auth/kyc/status/	Check KYC status	JWT
POST	/api/v1/auth/logout/	Invalidate tokens	JWT

Sample: Verify OTP Response

```
POST /api/v1/auth/verify-otp/
Body: { "phone": "+919876543210", "otp": "482931" }

Response 200:
{
  "access": "eyJhbGciOiJIUzI1NiIsInR5...",
  "refresh": "eyJhbGciOiJIUzI1NiIsInR5...",
  "user": { "id": "usr_01HX...", "kyc_status": "pending", "name": "Riya Sharma" }
}

Error 400: { "error": "OTP_EXPIRED", "message": "OTP has expired. Request a new one." }
Error 429: { "error": "RATE_LIMITED", "message": "Too many attempts. Try after 30 minutes." }
```

Card Module

GET	/api/v1/cards/	List all cards for user	JWT
GET	/api/v1/cards/{id}/	Card detail + balance	JWT
POST	/api/v1/cards/{id}/freeze/	Freeze card instantly	JWT
POST	/api/v1/cards/{id}/unfreeze/	Unfreeze card	JWT
POST	/api/v1/cards/{id}/controls/	Set spending controls	JWT
POST	/api/v1/cards/{id}/limit-request/	Request limit increase	JWT

Transaction Module

GET	/api/v1/transactions/	List with filters/pagination	JWT
GET	/api/v1/transactions/{id}/	Transaction detail	JWT
POST	/api/v1/transactions/{id}/dispute/	Raise dispute	JWT

POST	/api/v1/transactions/{id}/receipt/	Attach receipt image	JWT
POST	/api/v1/webhooks/card-network/	Receive txn from Visa/MC	HMAC Sig

Billing & Payments Module

GET	/api/v1/billing/statements/	List statements	JWT
GET	/api/v1/billing/statements/{id}/	Statement detail + PDF	JWT
POST	/api/v1/billing/pay/	Initiate payment	JWT
POST	/api/v1/billing/autopay/setup/	Configure auto-pay	JWT
POST	/api/v1/billing/emi/convert/	Convert txn to EMI	JWT

Standard Error Response Format

```
{
  "error": "ERROR_CODE",          // Machine-readable code
  "message": "Human friendly msg",
  "details": { ... },             // Optional field-level errors
  "request_id": "req_01HX..."     // For support tracing
}

HTTP Status Codes:
  200 OK          — Success
  201 Created     — Resource created
  400 Bad Request — Validation error
  401 Unauthorized — Token invalid/expired
  403 Forbidden   — Insufficient permissions
  404 Not Found   — Resource not found
  429 Too Many Requests — Rate limited
  500 Server Error — Internal error (logged + alerted)
```

06 Frontend Architecture (Mobile-First)

App Structure & Navigation

```
=== CREDIFY APP — SCREEN HIERARCHY ===

Auth Stack (unauthenticated)
  ├── Splash / Onboarding
  ├── Phone Entry + OTP
  └── KYC Flow (doc upload, liveness)

Main Tab Navigator (authenticated)
  ├── [Home] Dashboard
  │      ├── Spend Summary Widget
  │      ├── Card Widget (swipeable)
  │      ├── Recent Transactions (5)
  │      └── Quick Actions (Pay, Freeze, Rewards)
  ├── [Card] Card Management
  │      ├── Card Detail
  │      ├── Controls & Limits
  │      └── Request Physical Card
  ├── [Transactions] Full Transaction List
  │      ├── Filter / Search
  │      ├── Transaction Detail
  │      └── Dispute Flow
  ├── [Insights] Analytics
  │      ├── Spend by Category
  │      ├── Monthly Comparison
  │      ├── Budget Tracker
  │      └── Credit Score Widget
  └── [Profile] Account Settings
         ├── Personal Info
         ├── Notifications Preferences
         ├── Security Settings
         └── Support / Help

Modal Flows (overlaid)
  ├── Payment Flow (Bill Pay)
  ├── Rewards Redemption
  └── Support Chat
```

Key Screen Specs

Dashboard

- Available balance vs credit limit (donut/arc chart)
- Virtual card preview with tap-to-reveal card number
- Current statement amount + due date countdown
- Rewards points balance with tier badge
- Spend velocity: "You've spent 67% of last month's total"

Card Screen

- Flip animation for card front/back
- Freeze toggle with haptic feedback + confirmation
- Spending controls: toggles by merchant category
- Credit limit bar + request increase CTA
- Card history (issued, replaced dates)

Transaction Screen

- Infinite scroll with date group headers
- Filter chips: Date, Category, Amount, Status

- Merchant logo + enriched category icon
- Swipe-left gesture → quick dispute initiation
- EMI badge on converted transactions

Payments Screen

- Current due amount prominently displayed
- Pay minimum / Pay full / Pay custom — 3 clear CTAs
- Payment method selector with saved methods
- Auto-pay status card with edit option
- Payment history timeline

Insights Screen

- Animated category breakdown (donut chart)
- Month-over-month bar chart (last 6 months)
- Top 3 merchants by spend
- Budget progress bars per category
- Credit score trend line with factor explanations

UX Principles (Fintech-Focused)

- **Speed:** Critical actions (freeze, pay) < 2 taps from dashboard
- **Trust:** Always show masked card data; require biometric for full reveal
- **Clarity:** No hidden fees — always show fee before confirming payment
- **Accessibility:** WCAG 2.1 AA compliance; support screen readers
- **Offline:** Last cached data shown when offline; no blank screens
- **Micro-interactions:** Haptic feedback on card freeze, payment success animations

07 Analytics & Metrics

Events to Track

screen_view	**Every screen render**	**screen_name, user_id, session_id**
card_frozen	**User freezes card**	**card_id, reason, duration_ms**
payment_initiated	**User taps pay**	**amount, method, cycle_id**
payment_completed	**Gateway confirms**	**amount, method, latency_ms**
dispute_raised	**Dispute submitted**	**txn_id, amount, category**
reward_redeemed	**Points redeemed**	**points, type, value_inr**
kyc_completed	**KYC approved**	**provider, duration_mins, attempt_num**
fraud_alert_shown	**Fraud event triggered**	**txn_id, score, action**

insight_viewed	**Analytics tab open**	**category, month, time_spent**

Key KPIs

User KPIs

- DAU / MAU ratio (target: > 30%)
- Onboarding completion rate (KYC funnel)
- D1 / D7 / D30 retention
- Average session length
- Feature adoption rate per module

Financial KPIs

- Total Transaction Volume (TTV) month-on-month
- On-time payment rate (target: > 92%)
- Net interest income per user
- Rewards redemption rate (engagement signal)
- Dispute rate (target: < 0.5% of txns)
- Revenue per active user (RPAU)
- Cost of acquisition (CAC) vs Lifetime Value (LTV)

Admin Dashboard Metrics

- Real-time: active sessions, transactions per minute, fraud alerts queue
- Daily: new signups, KYC pass rate, payment collections
- Weekly: churn signals, support ticket SLA adherence
- Monthly: credit limit utilization distribution, revenue attribution

08 Smart Systems (Intelligence Layer)

Credit Scoring Engine

Phase 1 — Rule-Based (MVP)

- Weighted scorecard: payment history 35%, utilization 30%, bureau 20%, account age 15%
- Score range 300–900; refresh monthly or on significant event
- Risk bands: 750+ Low | 650-749 Medium | 550-649 High | <550 Very High

Phase 2 — ML-Based (V2)

- XGBoost model trained on bureau data + internal behavioral signals

- Features: velocity patterns, payment timing, category spend mix, device signals
- Model versioning with A/B shadow testing before promotion
- SHAP values for explainability (regulatory requirement)

Notification Rules Engine

```
Rule examples:

TRIGGER: txn.created
  IF amount > 10000 → SMS + Push (HIGH priority)
  IF merchant_category == GAMBLING → Alert + CRM flag
  IF txn outside_home_country → Push (with FX rate)

TRIGGER: billing.due_date_approaching
  IF days_remaining == 3 → Email + Push
  IF days_remaining == 1 AND balance > 0 → SMS + Push (URGENT)
  IF days_remaining == 0 AND unpaid → SMS + Push + WhatsApp

TRIGGER: fraud.score_high
  IF score > 0.8 → Freeze card + Multi-channel alert + CRM ticket
  IF score > 0.6 → Push alert + require re-auth for next txn
```

Fraud Detection

Rule-Based Triggers (real-time, < 200ms)

- Transaction > 5x user 30-day average → flag
- 3+ transactions within 60 seconds → velocity block
- New merchant country within 2 hours of domestic txn → flag
- Card-not-present transaction > ₹50,000 → require OTP confirmation

ML Layer (async, post-auth)

- Isolation Forest for anomaly detection on transaction features
- Outcome: score 0.0–1.0 → logged on fraud_alerts table
- High-score transactions reviewed by fraud team within 1 hour
- Model retrained weekly with new labeled data from dispute outcomes

09 Security & Compliance

Data Encryption

- At rest: AES-256 for PII fields (card numbers, PAN, Aadhaar)
- In transit: TLS 1.3 enforced; HSTS enabled

- Database-level field encryption for sensitive columns using django-encrypted-model-fields
- KMS (AWS KMS or HashiCorp Vault) for key management

Card Tokenization

- Primary Account Number (PAN) never stored in plain text
- Network tokenization via Visa Token Service / Mastercard MDES
- Internal token (card_number_token) mapped to vault reference only
- CVV stored only during transaction auth window, then purged

Audit Logs

- All mutations to sensitive entities logged in audit_logs table
- Log fields: actor_id, IP, action, before/after values, timestamp
- Logs are append-only (no UPDATE/DELETE on audit table)
- Retained for 7 years per RBI/financial regulation requirements
- Shipped to SIEM (Splunk / ELK) for real-time monitoring

KYC / AML Considerations

- KYC provider: Onfido (global) or CKYC (India) integration
- Document verification + liveness check mandatory for activation
- Re-KYC triggered for high-risk events (large limit increase, suspicious activity)
- AML screening: PEP/sanctions list check via Refinitiv World-Check
- Suspicious Activity Reports (SAR) workflow for compliance team
- PMLA compliance: transaction records retained for 5 years

PCI-DSS Awareness

⚠️ **PCI Scope Note**

Credify does NOT store full card numbers, CVV, or PAN directly.
Card processing is handled by a PCI-DSS Level 1 certified card processor (Visa/MC issuer processor).
Credify falls under PCI-DSS SAQ D (Service Provider) scope for tokenized data handling.
Annual penetration testing and quarterly ASV scans required.
Separate network segment for any cardholder data environment (CDE) components.

Additional Security Controls

- OWASP Top 10 mitigations enforced in code review checklist
- Secrets management: environment variables via AWS Secrets Manager (never in code)

- Rate limiting: per-IP and per-user on all auth endpoints
- CORS: strict whitelist of allowed origins
- Security headers: CSP, X-Frame-Options, X-Content-Type-Options
- Dependency scanning: automated via Snyk / Dependabot in CI pipeline

10 Product Roadmap

MVP	**Month 1–3**	**Core platform live**	**Auth + KYC, Virtual card, Transactions, Basic billing, Push notifications**
V1	**Month 4–6**	**Full feature set**	**Physical card, Rewards engine, Analytics, Disputes, CRM + AI chat**
V2	**Month 7–9**	**Intelligence layer**	**ML credit scoring, Fraud ML model, EMI, Budget tools, WhatsApp notif**
Scale	**Month 10–15**	**Growth + resilience**	**Multi-region, Partner APIs, Premium tiers, Credit Builder product, B2B card**

MVP Scope (Non-negotiable for launch)

- User registration + OTP auth
- KYC via document upload + liveness check
- Credit limit assignment (rule-based scoring)
- Virtual card issuance
- Transaction ingestion + real-time push notifications
- Monthly statement generation
- Basic payment (UPI / Net Banking)
- Card freeze/unfreeze
- Basic support ticket creation

V1 Additions

- Physical card issuance + tracking
- Rewards points engine + redemption
- Full analytics dashboard (category breakdown, trends)
- Dispute management workflow

- AI-powered support chat (first contact resolution)
- Auto-pay setup
- EMI conversion for eligible transactions

V2 Additions

- ML-based credit scoring (XGBoost)
- Real-time ML fraud detection
- Budget setting + proactive overspend alerts
- WhatsApp notification channel
- Referral program
- Merchant-funded offers

Scale Phase

- Multi-region deployment (AWS Mumbai + Singapore)
- Open API for fintech partner integrations (BNPL, insurance)
- Credit Builder secured card product
- Corporate / Business card variant
- ISO 27001 certification preparation

11 Business Model

Revenue Streams

Interchange fees	**% of every transaction processed (1.5–2% of GMV)**	**~50–60% of revenue**
Interest / Finance charges	**APR on revolving balance (24–42% p.a.)**	**~25–30% of revenue**
Late fees	**Flat fee for missed due date**	**~5% of revenue**
Annual / Joining fees	**Premium tier membership fee**	**~5% of revenue**
Partner merchant commissions	**Merchant-funded reward campaigns**	**~5–10% of revenue**
EMI processing fee	**1–2% origination fee on EMI conversions**	**~5% of revenue**

Cost Considerations

- Card network fees (Visa/MC): ~0.1–0.15% of GMV
- KYC provider: ₹80–150 per verified user

- Payment gateway: ₹2–5 per transaction
- SMS / notification costs: ₹0.15–0.20 per SMS
- Credit bureau pull: ₹35–60 per enquiry
- Cloud infrastructure: scales with user base (~₹200–400/user/year at scale)
- Customer acquisition cost (CAC): target ₹800–1,200 blended

Unit Economics (Indicative)

Avg credit limit	**₹1,50,000**	**—**
Avg monthly spend per user	**₹25,000**	**₹15,000/month**
Interchange per user/month	**₹375 (1.5%)**	**₹225**
Revolve rate	**35% of users carry balance**	**20% needed**
LTV (36 months)	**₹12,000–18,000**	**₹4,000+ to beat CAC**

12 Risks & Challenges

Technical Risks

Card network downtime	**Low**	**High**	**Circuit breaker, fallback queuing, SLA with issuer processor**
Kafka consumer lag spike	**Medium**	**Medium**	**Consumer group monitoring, auto-scaling, dead-letter queue**
Database lock contention	**Medium**	**High**	**Connection pooling (PgBouncer), read replicas for analytics**
ML model drift (fraud)	**Medium**	**High**	**Weekly retraining, performance monitoring, rule-based fallback**
Third-party KYC failure	**Low**	**High**	**Secondary provider fallback, manual KYC queue**

Compliance Risks

- RBI Master Directions on Credit Cards — strict adherence required for card issuance
- DPDP Act (India) — data privacy, consent management, right-to-forget implementation

needed
- AML / CFT — failure to detect suspicious activity = regulatory penalty + license risk
- KYC gaps — onboarding non-eligible users (minor, PEP) = severe compliance breach
- Interest rate disclosure — must be prominently displayed per RBI FPC guidelines

Scaling Challenges

- Transaction throughput: design for 10,000 TPS peak (card festival sales seasons)
- Statement generation: batch job for 100K+ users must complete in < 2 hours
- Notification fan-out: sending 500K push notifications within 30 seconds of event
- Cold start: building credit history for thin-file users without gaming risk models
- Fraud at scale: false positive rate must stay < 0.1% to avoid user friction

🔑 **Key Success Factors**

1. Nail the onboarding funnel — KYC drop-off is the #1 conversion killer in fintech.
2. Real-time experience is non-negotiable — any transaction delay > 3s erodes trust.
3. Build compliance as a competitive moat, not a checkbox.
4. Start with a narrow ICP (young professionals, metro cities) before expanding.
5. Revenue from interchange + revolvers — optimize for engaged, responsible borrowers.

— End of Document —

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